

Hye-Young Jo

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Research Interests

I am a Ph.D. student in Computer Science at the University of Colorado Boulder. My main research area is **Human-Computer Interaction**. My research goal is to **expand people's capacity to create and learn** by building interactive systems with generative AI and extended reality that adapt content to their actions, context, and intentions. My work spans AI-assisted filmmaking, technology-enhanced learning, and embodied fitness training.

Keywords: human-AI interaction, creativity support tools, adaptive media, embodied interaction

Education

University of Colorado Boulder

Ph.D. student, Computer Science
Supervisor: Ryo Suzuki

Aug 2024 - Present

Korea Advanced Institute of Science and Technology (KAIST)

Master of Science, Industrial Design, *Best Master's Thesis Award*
Supervisor: Andrea Bianchi

Sep 2020 - Aug 2022

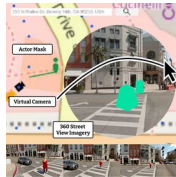
Seoul National University

Bachelor of Fine Arts, Painting and Media Arts, *Cum Laude*
Supervisor: Inhwon Oh, Cheol-Woong Sim

Mar 2011 - Feb 2016

Publication

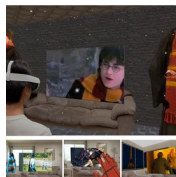
Summary: 8 full papers published in CHI (4), UIST (2), ISMAR (1), and HCIK (1), along with 1 extended abstract in UIST. One additional full paper is on arXiv.



[C8] Map2Video: Guiding Real-World-Grounded AI Video Generation

Hye-Young Jo, Mose Sakashita, Aditi Mishra, Ryo Suzuki, Koichiro Niinuma, Aakar Gupta

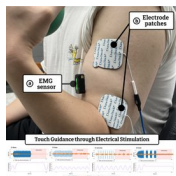
UIST 2026: ACM Symposium on User Interface Software and Technology (Acceptance rate: 20.1%) ([video](#)) ([PDF](#))



[C7] CinemaWorld: Generative Augmented Reality with LLMs and 3D Scene Generation for Movie Augmentation

Keiichi Ihara, DaeHo Lee, Manato Abe, **Hye-Young Jo**, Ryo Suzuki

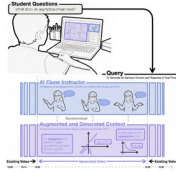
UIST 2026: ACM Symposium on User Interface Software and Technology (Acceptance rate: 20.1%) ([PDF](#))



[C6] TingleTouch: Touch Guidance through Electrical Stimulation in Resistance Training
Dong-Uk Kim, **Hye-Young Jo**, Hankyung Kim, Ryo Suzuki, Seungwoo Je, Yoonji Kim

CHI 2026: ACM Conference on Human Factors in Computing Systems (to appear)

(Acceptance rate: 25.3%) ([video](#)) ([PDF](#))



[P1] Generative Lecture: Making Lecture Videos Interactive with LLMs and AI Clone Instructors

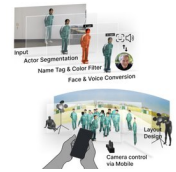
Hye-Young Jo, Ada Zhao, Xiaolan Liu, Ryo Suzuki
arXiv 2025 ([video](#)) ([PDF](#))



[C5] Designing Hand and Forearm Gestures to Control Virtual Forearm for User-Initiated Forearm Deformation

Yilong Lin, Han Shi, Weitao Jiang, Xuesong Zhang, **Hye-Young Jo**, Yoonji Kim, Seungwoo Je

ISMAR 2025: IEEE International Symposium on Mixed and Augmented Reality (Acceptance rate: 7.86%) ([video](#)) ([PDF](#))



[C4] CollageVis: Rapid Previsualization Tool for Indie Filmmaking using Video Collages
Hye-Young Jo, Ryo Suzuki, Yoonji Kim

CHI 2024: ACM Conference on Human Factors in Computing Systems (Acceptance rate: 26.4%) ([video](#)) ([PDF](#))



[E1] TrainerTap: Weightlifting Support System Prototype Simulating Personal Trainer's Tactile and Auditory Guidance

Hye-Young Jo, Chan Hu Wie, Yejin Jang, Dong-Uk Kim, Yurim Son, Yoonji Kim
UIST 2023 Adjunct: ACM Symposium on User Interface Software and Technology

(Acceptance rate: 21.0%) ([video](#)) ([PDF](#))



[C3] FlowAR: How Different Augmented Reality Visualizations of Online Fitness Videos Support Flow for At-Home Yoga Exercises

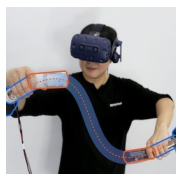
Hye-Young Jo, Laurenz Seidel, Michel Pahud, Mike Sinclair, Andrea Bianchi

CHI 2023: ACM Conference on Human Factors in Computing Systems (Acceptance rate: 27.6%) ([video](#)) ([PDF](#))



[C2] Design of Virtual Reality Application for Interaction Prototyping Remote Education
Hye-Young Jo, Wooje Chang, Hoonjin Jung, Andrea Bianchi

HCIK 2022: The Human-Computer Interaction Society of Korea ([video](#)) ([PDF](#))
Best Paper Award



[C1] GamesBond: Bimanual Haptic Illusion of Physically Connected Objects for Immersive VR Using Grip Deformation

Neung Ryu, **Hye-Young Jo**, Michel Pahud, Mike Sinclair, Andrea Bianchi

CHI 2021: ACM Conference on Human Factors in Computing Systems (Acceptance rate: 26.3%) ([video](#)) ([PDF](#))

Honorable Mention Award

Research Experience

Autodesk, HCI and Visualization Team

May 2026 - Aug 2026

Research Intern, Mentors: Frederik Brudy, David Ledo.

Created a design space for film directing, grounded in a cinematic corpus, to establish a shared language between human creators and generative AI.

Fujitsu Research of America, Converging Lab

May 2025 - Oct 2025

Research Intern, Mentors: Mose Sakashita, Aditi Mishra, Aakar Gupta, Koichiro Niinuma.

Built a system that grounds AI video generation in street-view imagery for spatially consistent shots, evaluated with 12 filmmakers [C8].

CU Boulder, Programmable Reality Lab

Aug 2024 - Present

Graduate Research Assistant, Mentor: Ryo Suzuki

Study how people create and consume video by building interactive systems with generative AI, extended reality, and virtual agents for storytelling and education [C7, C8, P1].

Chung-Ang University, Artifab Lab

Jan 2023 - Jul 2024

Research Assistant, Mentor: Yoonji Kim

Built a filmmaking previzualization tool using image segmentation and deepfake face swapping [C4], and haptic systems that recreate a personal trainer's touch through vibrotactile and electrical stimulation [E1, C6].

KAIST, Make Lab

Sep 2020 - Sep 2022

Graduate Research Assistant, Mentor: Andrea Bianchi

Designed and evaluated AR, VR, and haptic systems for home fitness, remote education, and immersive VR [C1-C3].

Fellowship and Awards

Summary: \$176,000 in total funding awarded.

Google Ph.D. Fellowship

Fall 2025 - Spring 2027

Full tuition and stipend (\$85,000/year)

CU Boulder Computer Science Department

2026

Conference Support funds (\$1,500)

Ralph J. Slutz Student Excellence Award

2025

Slutz Excellence Fund (\$1,000)

Graduate and Professional Student Government

2024

Conference travel support (\$500)

Gary Marsden Travel Award

2024

CHI conference travel support (\$3,000)

Best Master's Thesis Award

2023

Department of Industrial Design, KAIST

Best Paper Award

2022

HCIK 2022 [C2]

Top Research Award

2022

The joint research seminar of four universities: Seoul National University, KAIST, Sogang University, and Korea University [X6]

Excellence Award

2021

Korea Metaverse Developer Contest 2021 [X6]

IDEA Design Award 2021

2021

Bronze, Digital Interaction, Mobile Clinic Module Control UI/UX (awarded to Make lab)

iF Design Award

2021

Winner, User Interface (UI), User Experience (UX), Mobile Clinic Module Control UX/UI (awarded to Make lab)

Honorable Mention Award

2021

CHI 2021 [C1]

Top Award

2016

Green Computer Academy

Alumni Award

2016

Seoul National University Alumni Association

Work Experience

Film VFX Compositor and VR/AR Generalist at [Dexter Studios](#)

Sep 2016 - Apr 2020

Worked in film post-production's compositing team, integrating computer graphic assets, matte painting, and live-action footage to make a final image [F1-F6].

Created various AR/VR projects, mainly as a generalist, involved in 3D character design, level design, visual scripting, lighting, and look development [X1-X2, X4].

Service

Reviewer

CHI 2025-2026, DIS 2026

Outreach and Leadership

CU Boulder Prospect Affiliate Match Program, helping prospective students prepare their PhD applications 2025

Seoul National University Mentoring Program, helping international students adjust to campus life 2012

Invited Talks and Lectures

Fujitsu Research of America Lunch Seminar

Oct 2025

Grounding Generative Video in Real-World Geographies for Spatial Consistency

KAIST Make Lab Seminar

May 2025

Reimagining How We Create and Consume Content

Seoul National University Human-Centered Computing Systems Lab Seminar

May 2025

Reimagining How We Create and Consume Content

CU Boulder CSCI 7000 How to Hack Almost Anything Class Guest Lecture

Feb 2025

Utilizing AI Services and APIs for Coding and Virtual Agent Creation

KAIST ID220 Interaction Prototyping Class Guest Lecture

Oct 2022

DIY Arduino paper cases without 3D printing, using Adobe Illustrator, Autodesk Fusion 360, and Blender.

Kookmin University Department of Entertainment Design Career Seminar

Nov 2019

Transitioning from Art and Film VFX to VR Content Creation

Media Coverage

CU Boulder Graduate School News

Oct 2025

["CU Boulder graduate student named a Google PhD fellow"](#)

Microsoft Research Blog

May 2021

["Microsoft Research collaborates with KAIST in Korea to explore bimanual interactions with haptic feedback in virtual reality"](#)

Teaching and Mentoring

Teaching Assistant

CSCI 1300 Starting Computing (C++), Instructor: Rhonda Hoenigman

Fall 2025

ID220 Interaction Prototyping (Arduino)

Fall 2022

Undergraduate Research Assistant

Chan Hu Wie, Yejin Jang, Dong-Uk Kim, Yurim Son (Chung-Ang University)

Film and XR Content

Summary: I created traditional 2D films and 3D XR content as VFX Compositor [F1-F6], CG generalist [X1-X4, X6], Art director [X5], developer [X3], and camera operator [X4].

Films

[F6] <Along with The Gods: The Last 49 Days>, Director: Yong-Hwa Kim, 2018, *2018 Grand Bell Awards - Best Visual Effects, over 12 million viewers in South Korea alone as of 2024.*

[F5] <1987: When The Day Comes>, Director: Jun-Hwan Jang, 2017, *2018 Blue Dragon Film Awards - Top Film Award, over 5 million viewers in South Korea alone as of 2024.*

[F4] <Along with The Gods: The Two Worlds>, Director: Yong-Hwa Kim, 2017, *2018 Blue Dragon Film Awards - 4 Crowns, including Technology Award, over 11 million viewers in South Korea alone as of 2024.*

[F3] Kung Fu Yoga, Director: Stanley Tong, 2017

[F2] <Real>, Director: Sa-Rang Lee, 2017

[F1] <Fabricated City>, Director: Kwang-Hyun Park, 2017

Immersive Content

[X6] VR Boxing Game <Meta-Boxing> ([video](#)), Supervisor: Woontack Woo, *Excellence Award at Korea Metaverse Developer Contest 2021, Top Research Award at the 2022 joint research seminar of four universities - Seoul National University - KAIST - Sogang University - Korea University*

[X5] VR Exhibition <Being City>, Supervisor: Tae-Kyung Yoo

[X4] AR Mobile App <LGU+ 5G AR>, Supervisor: Sun-Gu Kim

[X3] VR Exhibition <Fashion For Help>, Supervisor: Young-Mo Son

[X2] AR Exhibition <The Tide>, Supervisor: Sang-Hyoun Lee

[X1] VR Toon Film <The Tide>, Director: Tae-Kyung Yoo, *Officially invited to the "New Frontier" category at the 2019 Sundance Film Festival.*

Art Exhibitions and Live Performances

Summary: I participated in the following art events as an artist [A1-A5, A7] and VJ [A6].

[A7] Group Exhibition <Lapses>, Platform-L Contemporary Art Center, Curator: Eobchae, 2018 (funded by Hyundai's ZERO1NE project).

[A6] Live VJ Show <We Play>, Sangsangmadang, Supervisor: Hoon-Gyu Park (Parkpunk), 2016

[A5] Invited Exhibition <Don't fake it, believe it>, Mythtake Museum, 2016

[A4] Group Exhibition <Dirt Luv for Graduation>, Seoul National University, 2016

[A3] Group Exhibition <8-bit>, Seoul National University, 2015

[A2] Group Exhibition <Crawling>, Seoul National University, 2015

[A1] Group Exhibition <The Great Exodus>, Seoul National University, 2015

Skills

Programming

C#, Python, JavaScript, C++, R, HTML/CSS

Graphic Design

2D/3D image manipulation (Nuke, Adobe Creative Suite), video editing (Final Cut Pro), 3D modeling, rigging, animation (Blender, Maya, Fusion 360, Substance Painter), projection mapping (Resolume), live video mixing (VDMX)

XR Prototyping

Unity, Unreal Engine, WebXR

Fine Art

Drawing, painting (watercolor, acrylic, oil), sculpture, casting, printmaking, photography

Research Methods

affinity diagramming, A/B testing, focus group interviews, ethnography, contextual inquiry, participatory design

Languages

Korean (native), English (TOEFL 106/120), French (DELF B1)